TITLE OF THE REPORT SHOULD BE WRITTEN HERE IN ARIAL NARROW, 18 PT, BOLD, ALL CAPS, AND CENTER-ALIGNED

by

NAME OF STUDENT 1

NAME OF STUDENT 2

BACHELOR OF SCIENCE
IN
ELECTRICAL AND ELECTRONIC ENGINEERING

Department of Electrical and Electronic Engineering
INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

MAY 2026

NAME OF THE TITLE IS HERE, TIMES NEW ROMAN BOLD, 18 PT CENTRE JUSTIFIED

by

NAME OF STUDENT 1

NAME OF STUDENT 2

A thesis/project
submitted as partial fulfilment of the requirement for the degree of

BACHELOR OF SCIENCE
IN
ELECTRICAL AND ELECTRONIC ENGINEERING

Department of Electrical and Electronic Engineering
INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

MAY 2026

CERTIFICATE OF APPROVAL

The thesis/project entitled "**Name of your Thesis/Project Title**" submitted by **Name of Student 1** (Matric ID: **ET..)** and **Name of Student 2** (Matric ID: **ET...)** of the session Spring 2022, to the Department of Electrical and Electronic Engineering, International Islamic University Chittagong, has been accepted as satisfactory in partial fulfilment of the requirements for the degree of Bachelor of Science in Engineering and approved at the examination held on **May 9, 2026**.

Note: Session refers to the intake period of a student. For example, Student ID ET191001 belongs to the Spring 2019 session, while ET223023 corresponds to the Autumn 2023 session.

Date of examination refers to the date on which the final defense presentation is conducted.

Signature of the supervisor

Name: **(**Supervisor**)**
Designation:
Department of Electrical and Electronic Engineering
International Islamic University Chittagong

Omit this section (co-supervisor signature and information) if there is no co-supervisor.

Signature of the co-supervisor

Name: **(co-s**upervisor**)**
Designation:
Department of Electrical and Electronic Engineering
International Islamic University Chittagong

DECLARATION

I/We hereby declare that this thesis/project report is the result of my/our own independent work and that no part of this work has been submitted elsewhere for the award of any degree or diploma. This work does not include any contribution from collaborative work, except where explicitly stated in the text.

I/We further declare that:

Originality and Plagiarism

This report is an original work. All sources of information, ideas, and data have been properly acknowledged and referenced in accordance with the guidelines of the Department of Electrical and Electronic Engineering (EEE), International Islamic University Chittagong (IIUC).

Data Integrity

The data presented in this report are genuine, and have been collected, processed, analyzed, and reported with academic honesty. No data have been fabricated, falsified, or manipulated in any form to influence the results or conclusions.

Human/Animal Subjects *(select as applicable)*

☐ This research did not involve human or animal subjects.

☐ This research involving human participants was conducted with prior ethical approval from the relevant authority. Informed consent was obtained from all participants, and their confidentiality has been strictly maintained.

Conflict of Interest

The authors declare that there is no conflict of interest that could have influenced the conduct, analysis, or interpretation of this study.

Ethical Compliance

This work has been carried out in accordance with the research ethics guidelines instructed by the department.

Name of the Students	**Signature of the Students**
Name of Student 1	Signature of Student 1
Name of Student 2	Signature of Student 2
Name of Student 3	Signature of Student 3

ACKNOWLEDGMENT

(This is a sample acknowledgment section intended for reference. Students are encouraged to modify it according to their specific requirements.)

All praise and thanks to Allah, the Lord of the worlds, the Most Beneficent, the Most Merciful for granting us the ability to accomplish this work.

We would like to express our sincere gratitude to all those who supported and guided us throughout the completion of this thesis/project. We are especially thankful to our supervisor for his/her valuable guidance, encouragement, and continuous support.

We also extend our appreciation to our teachers, family members, and friends for their assistance and motivation during this work.

Authors

ABSTRACT

The abstract should provide a clear and concise overview of the research work, beginning with a brief introduction that outlines the significance, background, and motivation of the study. It should help the reader understand why the research was conducted and what problem or gap it addresses. After the introduction, the abstract should summarize the key components of the study, including the main objectives, the methodology or approach used, and the most important findings or results. Whenever possible, quantitative results should be included to make the outcomes more precise and meaningful. This helps readers quickly grasp the effectiveness and impact of the research. The concluding part of the abstract should highlight the major implications, significance, and contributions of the work. It should clearly state what the study adds to the existing body of knowledge and how the results may be applied or extended in future research or practical applications. An abstract must be self-contained, meaning it should be understandable without referring to the full thesis or report. The recommended structure is a single paragraph; however, two paragraphs may be used if necessary for clarity and flow. The total length should generally be within 300–400 words. It is important to maintain focus and avoid unnecessary details. Citations, references, figures, tables, block diagrams, and bullet points should not be included in the abstract.

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF ABBREVIATIONS

(All abbreviations should be listed in alphabetical order.)

AAS	Atomic Absorption Spectrophotometer
CNT	Carbon Nanotube
NP	Nanoparticle

CHAPTER 1
INTRODUCTION

1.1 Background

The introduction section of a thesis should provide the overall foundation and direction of the study. It should begin with the **background of the study**, where a general overview, context, and description of the research field are presented, along with relevant existing developments related to the problem area. This should be followed by the **problem statement**, in which the specific issue, research gap, or limitation in the current knowledge or system is clearly identified to justify the need for the study.

A separate subsection titled **significance of the study** should be included to explain why the research is important and how it contributes to addressing real-world challenges as well as advancing academic or technical understanding in the relevant field. After that, the **scope of the work** should be described by clearly defining the boundaries of the research, specifying what is included and excluded, preferably in both national and global contexts to ensure clarity and focus.

Finally, the section should include **thesis organization**, where a brief chapter-wise outline of the entire thesis is provided. This should help the reader understand the structure and logical flow of the document from the introduction through to the conclusion.

The entire thesis should be written in **Times New Roman** font, with **Justified** alignment, **1.5** line spacing, and a font size of **12 pt**. The text should not be bold unless required for headings. The page size must be **A4**, with margins set as follows: **Top=1"**, **Bottom=1"**, **Left=1.5"** and **Right=1"**.
The introduction section should have a background, problem statement, Significance of the work in national or global context, Scope of the work, Thesis organization.

1.2 First level heading (Times New Roman font, Bold and a font size of 12 pt.)

1.2.1 Second level heading (Times New Roman font, Bold, Italic and font size: 12)

1.2.1.1 Third level heading (Times New Roman font, Italic, and a font size of 12 pt.)

2.1 Guidelines for figures and tables

Every figure and table must include a clear caption and be properly referenced in the main text. For example, a figure should be introduced as: **"The components of a typical DSC are shown schematically in Fig. 1.1."** Similarly, each table and figure should be cited at an appropriate point in the discussion to ensure proper flow and clarity.

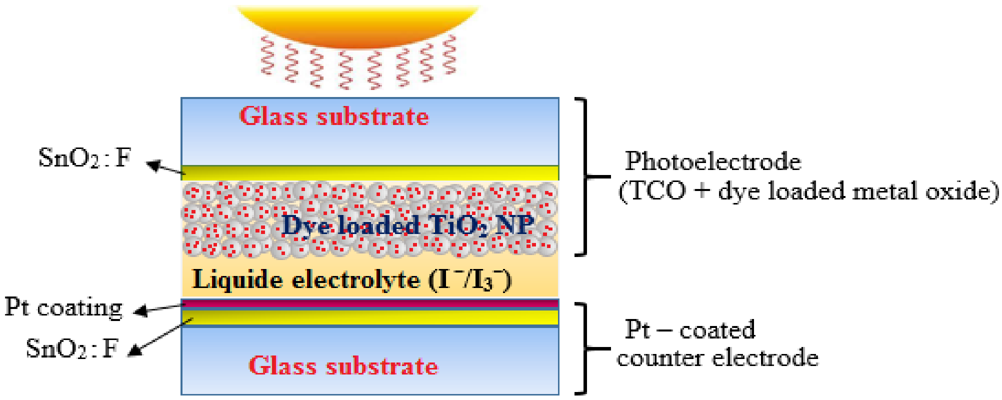

Fig. 1.1 Schematic of a simple DSC structure

The caption of a figure should be placed below it (e.g., **Fig. 1.1 Schematic of a simple DSC structure**), while the caption of a table should be placed above the table. All figures and tables should be centered within the printable area and must remain within the page margins.

Ensure that all figures and tables are clear, readable, and properly formatted. Text within figures should be legible and consistent with the thesis body font size. Figures and tables must be center-aligned and placed within the page margins. Maintain appropriate resolution and aspect ratio (height–width) for all figures. If any figure or table is reproduced from another source, it must be properly cited in the caption. Table captions should be placed above the table. Overall, all figures and tables should be presented in a neat, consistent, and professional format that enhances the readability and quality of the thesis.

2.3 Citation – Brief Instruction

All sources of information used in the thesis must be properly cited in the text using a consistent referencing style (e.g., IEEE format). Citations should be placed within square brackets and numbered sequentially according to their first appearance in the text.

Example of citing a reference

Burning of the fossil fuels causes a huge quantity of CO2 which is known as greenhouse gas, is responsible for of global warming [1].

Example of citing two references

The generation of electricity at dye-sensitized oxide electrodes in electrochemical cells was discovered in the late 1960's [7], [8].

Example of citing more than two references

However, the overall power conversion efficiency could not exceed over 2% due to the poor charge-collection efficiency from the dye-molecules [9]-[12]. or The band gap, composition, morphology, and thickness of the metal oxide layer influence the charge collection, transport, and light harvesting properties [10], [12]-[15]

Ensure that every cited reference appears in the reference list and that all references in the list are cited in the text. Citations should be placed at appropriate points in the sentence, typically at the end of the relevant statement. Consistency and accuracy in citation style must be maintained throughout the thesis.

CHAPTER 2
LITERATURE REVIEW

The Literature Review chapter should present a critical and organized summary of previous research related to the topic of the thesis. It should begin by reviewing relevant existing studies, theories, methods, and technologies that are directly connected to the research area. The discussion should not be merely descriptive; instead, it should compare, analyze, and synthesize the findings of different works to show their strengths, limitations, and relevance to the current study.

This chapter should clearly identify the **research gap** by highlighting what has already been done and what still remains unresolved or insufficiently addressed in the existing literature. Based on this gap analysis, the **motivation and justification** for the present research should be clearly established. The chapter should also lead naturally toward the formulation of the **research objectives**, showing how the proposed work is intended to address the identified gap.

The writing should be well-structured, logically connected, and focused on the specific problem of the thesis rather than a general discussion of the field. Proper citations and references must be used throughout to support the discussion and acknowledge original sources.

CHAPTER 3
SYSTEM COMPONENTS AND METHODOLOGY/SYSTEM DESIGN AND METHODOLOGY EXPERIMENTAL METHODOLOGY/MATERIALS AND METHODS

3.1. Introduction:

Students are advised to select an appropriate chapter title based on their research field. However, this chapter must include at least two sections: (i) **System and Components** (or **System Components and Resources / Materials and Instruments / Software and Components**) and (ii) **Experimental Methodology** (or **Methodology**), depending on the nature of the research. Additional sections or subsections may be included as required by the scope and type of the study

3.2. System and components:

Systems and components is a mandatory part/section of the thesis. This section shall describe the hardware and/or software tools used in the development of the proposed system and explain how these tools are selected and applied to address the research problem effectively.

The section shall provide a clear description of each relevant component, its functional role in the system, and its contribution toward achieving the research objectives. Where applicable, the selection of tools shall be justified by comparing alternatives and explaining the reasons for choosing the final implementation.

However, **do not include detailed descriptions of commonly known Electrical and Electronic Engineering components and instruments**, such as basic electronic components, microcontrollers, wires, multimeters, oscilloscopes, or similar standard laboratory equipment, as these are considered fundamental knowledge.

A brief discussion may be included only for **interdisciplinary, advanced, or less common components or tools** that are not typically familiar to Electrical and Electronic Engineering students and are specifically relevant to the proposed work.

The section must also identify and discuss the **limitations of the selected hardware and/or software tools**, highlighting practical constraints that may affect system performance or implementation. This discussion should demonstrate the ability to evaluate engineering solutions in the presence of real-world limitations, in accordance with **Program Outcome (PO-e)** requirements.

3.3 Experimental and Methodology/Methodology

This section will clearly describe the procedures, techniques, and approaches adopted to carry out the research, whether the work is based on simulation, experimental investigation, or a combination of both. This chapter should present a systematic and logical framework demonstrating how the research problem is addressed and how the objectives are achieved.

For **simulation-based work**, it should describe the modeling process, assumptions, governing equations (if applicable), software tools used, and simulation parameters. For experimental work, it should detail the experimental setup, instruments, materials, procedures, and operating conditions. In hybrid cases, the integration between simulation and experimental validation should be clearly explained.

A step-by-step description of the **methodological workflow** shall be provided, including system design, data collection, processing, and analysis techniques. The procedures should be described with sufficient clarity to ensure transparency and reproducibility of the study.

It is strongly recommended to use **block diagrams, flowcharts, circuit diagrams, and other relevant figures** to illustrate the methodology or experimental setup, as these enhance clarity and understanding.

The chapter should be well-structured and sufficiently detailed so that another researcher can replicate the work based on the provided description.

CHAPTER 4
RESULTS AND DISCUSSION

4.1 General Guideline on this chapter:

The Results and Discussion chapter shall present the findings of the study in a clear, logical, and well-organized manner, followed by a critical interpretation of those results. It must explain what the results mean and how they relate to the research objectives.

The results should be presented using appropriate formats such as tables, graphs, figures, or charts. All results must be properly labeled, captioned, and referred to in the text. For simulation-based studies, include relevant output plots, performance curves, or comparative results. For experimental work, present measured data, observations, and processed results. Ensure that the data presented are clear, accurate, and directly relevant to the objectives of the study.

The discussion should analyze and interpret the results in depth. It should explain trends, relationships, and key observations, and compare the findings with expected outcomes, theoretical predictions, or previously published works where applicable. Any deviations, limitations, or unexpected results should be clearly explained with proper reasoning.

Where relevant, include comparative analysis (e.g., with existing methods, models, or systems) to demonstrate the effectiveness and contribution of the proposed work. The discussion should highlight how the results address the research problem and fulfill the stated objectives.

Avoid presenting raw data without explanation. Ensure a strong connection between results and discussion, maintaining clarity, coherence, and logical flow throughout the chapter.

4.xx Assess Issues: societal, health, legal and cultural

Should consider societal, health, safety, legal and cultural issues in design (for PO-f). Students must include a section titled ‘Assess Issues: societal, health, legal and cultural’ in the ‘Results and Discussion’ chapter. In this section, students should assess

/evaluate their proposed solution within a real-world context, taking into account local conditions, user requirements, and applicable standards.

For example, students may discuss safety risks (e.g., electric shock, short circuit or fire hazards, battery explosion risks, overcurrent and overheating protection, and safe enclosure and insulation), health concerns (e.g., radiation exposure, electromagnetic field effects, overheating-related health hazards, eye strain from prolonged display use, and psychological effects), societal benefits (e.g., improved efficiency, accessibility, low-cost solutions, and employment impact), legal requirements (e.g., compliance with standards such as IEC, IEEE, or BSTI; use of licensed software/tools; data privacy; and certification requirements), and cultural considerations (e.g., ease of use for local users, language of the interface, user habits and behavior, and gender and accessibility considerations). "Gender or accessibility considerations" means designing and evaluating your engineering solution so that it is usable, safe, and inclusive for all types of users, regardless of gender, physical ability, or other limitations.

4.xy Sustainability, environmental, and societal impacts

For PO-g, students should include a section in the Results and Discussion chapter to evaluate the sustainability, environmental, and societal impacts of the proposed engineering solution.

Environmental impact may include analysis of energy consumption, material/component usage, pollution or emissions, and waste generation (e.g., e-waste, batteries).

Sustainability considerations may include long-term usability, maintainability and scalability, the use of energy-efficient or reusable components, and lifecycle thinking (design → use → disposal). Maintainability refers to designing the system so it can be easily repaired, updated, or improved over its lifetime. Scalability is the ability of a system to handle increased load, size, or scope without major redesign.

CHAPTER 5
CONCLUSION

The **Conclusion** chapter shall summarize the overall outcomes of the research in a clear and concise manner. It should restate the main objectives of the study and briefly highlight how these objectives have been achieved based on the results and analysis presented in previous chapters.

This section should emphasize the **key findings and contributions** of the work without repeating detailed explanations or data. It should clearly indicate the significance of the results and how the study addresses the identified problem or research gap.

The conclusion should also include a brief discussion of the **limitations of the study**, acknowledging any constraints, assumptions, or factors that may affect the generalization or performance of the proposed work.

Finally, the chapter should provide **recommendations for future work**, suggesting possible improvements, extensions, or further research directions that can build upon the present study.
The conclusion must be concise, focused, and directly derived from the work presented in the thesis. No new data, analysis, or references should be introduced in this section.

REFERENCES

Do not assign a chapter number to the references section; simply use the heading **REFERENCES**. The heading should be in **14 pt** font size, while the reference text should be in **10 pt** font size.

All references should be formatted according to the IEEE citation style, as outlined in the guidelines provided at the following link.

https://ieee-dataport.org/sites/default/files/analysis/27/IEEE%20Citation%20Guidelines.pdf

Important Rules

- **References must be numbered in order of appearance in the text**
- **Always ensure every citation in text appears in reference list**
- **Do not mix IEEE style with other styles**
- **Keep formatting consistent (font, spacing, punctuation)**

Format for book

[1] A. A. Author, Title of the book (in italics), Edition, Volume (if a multivolume work), Place of publication: Publisher, Year, page number(s) (if appropriate).

Example for book

[1] W. K. Chen, *Linear Networks and Systems,* 2nd Edition, Volume II, New York: McGraw-Hill, 1993, pp. 123-135.

[2] …….

Format for Journal

[#] A. A. Author of article. "Title of article", *Title of Journal*, Vol. #, no. #, pp. page number/s, year.

Example for Journal article

[1] E. P. Wigner, M. Gräzel, and G. Liu, “Theory of traveling wave optical laser”, *Material Research Bulletin,* Vol. 134, pp. 120-130, 1965.

[2] …….

Format for Conference

[#] Authors, "Title of article", *Name of the conference*, Venue, date, year.

Example for conference

[1] M. A. Choudhury, “Arsenic removal from water using graphite”, *National conference on Physics of Technology*, IIUC, April 10, 2026.

[2] …….

Format for Internet link

[#] Name of the topic, date of excess, retrieved from: Internet link.

APPENDIX A

It is not compulsory to include additional appendix. However, if you include, the font size would be **10** point, **Times New Roman**, with a line spacing of **1.15**.

APPENDIX B

The last two Appendixs should be justification of the work as a complex engineering problem and complex engineering activities. Must cite the heading where the evidences are incorporated in the report for each WP's (for PO-b). The font size would be **10 pt**, **Times New Roman**, with a line spacing of **1.15**.

Range of Complex Engineering Problem Solving

Name of Complex Engineering Attribute	Explanation (How addressed?)
P1 (Depth of knowledge required)	
P2 (Range of conflicting requirements)	
P3 (Depth of analysis required)	
P4 (Familiarity issues)	
P5 (Extent of applicable codes)	
P6 (Extent of stakeholder involvement and conflicting requirements)	
P7 (Interdependence)	

APPENDIX C

Complex activities mean (engineering) activities or projects that have some or all of the following characteristics. A justification of the complex engineering activities (WA's) that are connected to your thesis/project is to provide here. Must cite the heading where the evidences are incorporated in the report for WA's (For PO-j). The font size would be **10** point, **Times New Roman**, with a line spacing of **1.15**. It is not essential to show the attainment of all the attributes of Complex Engineering Activities.

Range of Complex Engineering Activities.

Attributes	**Explanation (How addressed?)**
A1 (Range of resources)	
A2 (Level of Interaction)	
A3 (Innovation)	
A4 (Consequences for society and the environment)	
A5 (Familiarity)	